

Plotly → Dynamic graph

Visualisation → Matplotlib &

Interactive → better storytelling

↳ seaborn
↳ Static graph

dashboards, Analytical tools

Relationship b/w two input features → Scatterplot
(continuous)

$f_1 \uparrow$ $f_2 \uparrow$

↳ positively correlated

$f_1 \uparrow$ $f_2 \downarrow$

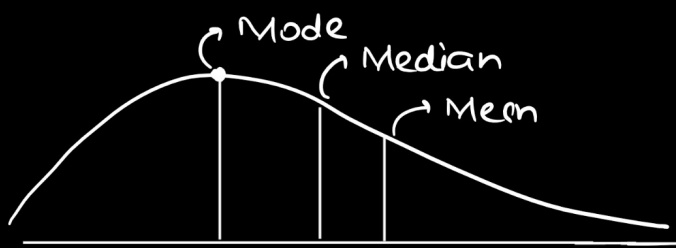
↳ negatively correlated

↳ no correlation

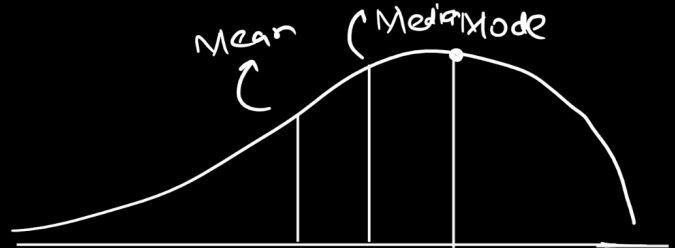
trends (upward/downward) → Line plot

Categorical data wrt continuous → bar plot

Distribution of any column → histogram



Right skewed



Left skewed

Mode < Median < Mean

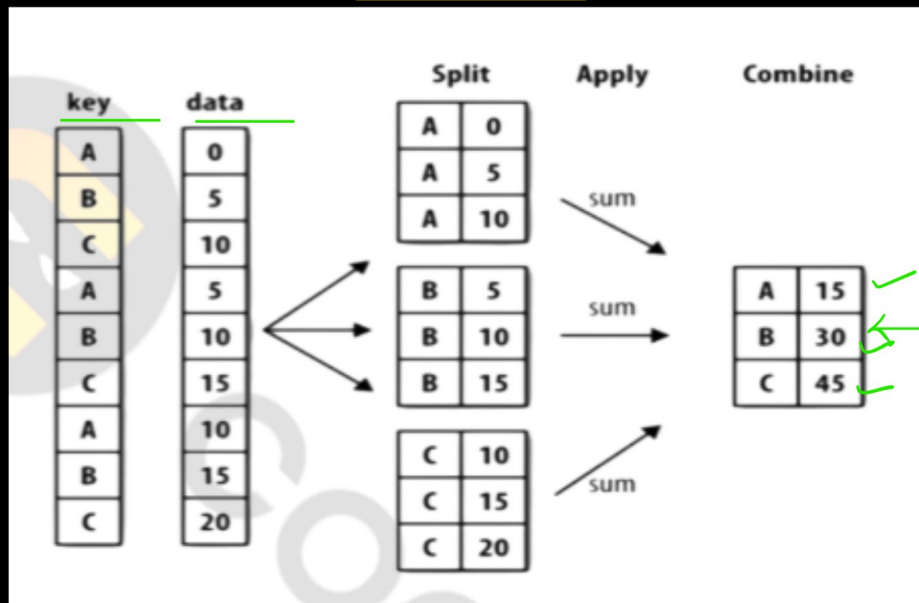
Mode > Median > Mean

Outliers

to detect the
Outliers

→ box/violin plot

Group by



Pivot table → Summarization

Column → Product A B C

Index

Region

East

West

Products

sum

Sales

✓

✓

✓

index —————
2026

Column	DA	DSA	DS
Jan			
Feb			
March			

Pivot-tables

Crosstabs → Count the frequency
of the categorical
data

↖ Wide format to a Long format
Melting → Wide format vs Long format

Subject	X	XOrd	Y
1	Scroll	2	152
1	Search	1	98
1	Voice	3	86
2	Scroll	1	148
2	Search	2	57
2	Voice	3	65
3	Scroll	2	160
3	Search	1	86
3	Voice	3	87
4	Scroll	1	113
4	Search	2	65
4	Voice	3	75
...	...	...	...

WIDE FORMAT

Common in stats packages for repeated measures ANOVAs.
Not much in R, but sometimes.

Subject	Scroll	Search	Voice	1	2	3
1	152	98	86	98	152	86
2	148	57	65	148	57	65
3	160	86	87	86	160	87
4	113	65	75	113	65	75
...	...	...	...	...	...	...

LONG FORMAT

Most common for R analyses.
Easiest to write to log files after each TRIAL.